Edge Vision: Real-Time Skeleton-Based Violence Detection using ST-GCN and Multi-Agent Kalman Tracking

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Abstract*—Real-time automated violence detection across distributed surveillance cameras is constrained by hardware compute ceilings, high false-positive rates on benign motion, and privacy hazards when raw video is streamed to centralized servers. Traditional 3D convolutional and video-transformer architectures operate over dense RGB pixel volumes, yielding large memory footprints and brittle sensitivity to background clutter, illumination shifts, and camera jitter. This work introduces Edge Vision, a dual-tier real-time violence detection platform unifying lightweight pose estimation (YOLOv8n-Pose), a Spatial-Temporal Graph Convolutional Network (ST-GCN), an 8-state constant-velocity Kalman occlusion tracker with Hungarian bipartite assignment, and a novel pixel-space geometric Relational Strike Gate. Representing each subject via normalized 17-node COCO skeletal coordinates across 30-frame temporal windows achieves over 99.9% per-frame, per-actor input dimensionality reduction (reducing single 1920×1080×3 RGB frames of 6.22 MB to 17×2 32-bit floats [136 bytes] per actor for ST-GCN, parsed from 17×3 pose triplets, while the downstream ST-GCN classifier relies on a 30-frame temporal graph tensor of shape (1, 2, 30, 17) for final sequence classification) while decoupling biomechanical motion from appearance confounders. The identical ONNX graph pipeline runs inside both a memory-bounded FastAPI REST cloud microservice (<512 MB RAM ceiling) and a zero-upload client browser Edge Mode accelerated via WebGPU and COOP/COEP-isolated multi-threaded WebAssembly (ONNX Runtime Web). Benchmarked over 100 timed iterations (after 10 warm-up runs) including OpenCV decoding and YOLOv8n-Pose extraction on an 11th Gen Intel Core i5-1135G7 CPU, strided execution (pose_stride=2, 4 threads) achieves 22.04 ± 14.87 ms/frame (Median P50: 24.05 ms, P95: 48.52 ms; 45.4 processed-frame throughput (FPS) defined as 1000/mean; 16.32 ms / 61.3 FPS process-only). Because pose_stride=2 is used, this represents the model's processing speed rather than the source-video ingestion rate. On a held-out test split of 293 unseen videos partitioned via an ordered source-ID split from 1,951 verified RLVS clips to prevent source-level leakage (TP=146, FP=46, FN=4, TN=97), the complete system achieves 82.94% Accuracy, 76.04% Precision, 97.33% Recall, 67.83% Specificity, and an F1 Score of 0.854.

***Index Terms*—Violence detection, spatial-temporal graph convolutional networks, pose estimation, Kalman filtering, Hungarian algorithm, relational strike gating, WebAssembly, edge AI.**

I. INTRODUCTION

Automated surveillance systems deployed in public transit corridors, campuses, correctional facilities, and retail spaces require sub-second detection of physical altercations while minimizing operator alert fatigue. Transmitting continuous high-definition video streams to centralized GPU clusters incurs prohibitive network egress costs, variable transmission latency, and regulatory compliance risks when personally identifiable visual frames leave local premises. Conversely, deploying heavy deep learning detectors directly onto low-power edge nodes, browser sandboxes, or memory-capped cloud micro-containers imposes strict limits on RAM consumption, binary size, and single-frame compute latency.

First-generation deep learning violence detectors processed volumetric RGB pixel tensors directly using 2D+LSTM hybrids or 3D Convolutional Neural Networks (3D-CNNs) such as C3D [1]. Although effective on curated indoor clips, dense pixel models exhibit fundamental operational drawbacks: (i) cubic memory scaling over spatial height, spatial width, and temporal depth, rendering real-time inference infeasible without high-wattage discrete GPUs; and (ii) severe confounding by nuisance visual factors including lighting flicker, rain, camera vibration, and clothing color.

Skeleton-based action recognition mitigates both challenges by projecting each human subject into a compact topological graph of anatomical keypoints [2]. Compressing 1920×1080×3 RGB arrays (6,220,800 channel bytes per frame) into normalized (17, 2) 32-bit float graph coordinates (136 bytes per actor for ST-GCN, parsed from 17×3 pose triplets of 204 bytes) achieves over 99.9% per-frame, per-actor input dimensionality reduction while stripping background texture and facial identity cues prior to sequence modeling (note that while per-frame inputs are reduced to 136 bytes per actor, the ST-GCN classifier relies on a 30-frame temporal graph tensor (1, 2, 30, 17) for final sequence classification). However, standalone skeletal graph classifiers remain vulnerable to two failure modes in unconstrained surveillance feeds: (1) track fragmentation and identity swaps during physical grapples where combatants heavily occlude one another, and (2) elevated false-positive rates caused by rapid isolated movements—such as dancing, vigorous gesturing, or solo jumping—that exhibit high joint velocity without hostile inter-actor contact.

To resolve these failure modes while preserving real-time CPU throughput, we present **Edge Vision**, making four concrete technical contributions:

1. **Unified Missing-Joint Masked ST-GCN Pipeline:** A lightweight residual Spatial-Temporal Graph Convolutional Network exported as a single self-contained 257 KB ONNX binary paired with mid-hip root centering, torso scaling, and strict [0.0, 0.0] validity masking for occluded joints.
2. **True 8-D Kalman + Hungarian Occlusion Tracker:** Analytic constant-velocity Kalman state prediction and optimal IoU linear assignment; qualitative observation during validation indicates the 8-D state vector preserves track identities through transient occlusions, reducing ID switches compared to greedy IoU matching.

3. **Pixel-Space Relational Strike Gating:** A deterministic dual-invariant geometric filter measuring minimum Euclidean effector-to-target-zone proximity (d_{strike} in pixels) alongside bounding-box overlap (IoU > 0), explicitly addressing edge geometries where naive box-centroid distance heuristics fail.
4. **Hybrid Browser Edge Mode & Cloud REST Architecture:** Browser WASM inference successfully reproduces the expected classification outputs on two regression test clips, demonstrating functional parity with the Python backend and guaranteeing zero-upload client privacy via `COOP/COEP SharedArrayBuffer` WASM SIMD and WebGPU, operating alongside a standard 512 MB FastAPI Cloud REST API.

II. RELATED WORK

Research in automated violence detection and skeletal kinematics has evolved rapidly between 2020 and 2026 across four intersecting subfields.

A. RGB-Volumetric and Weakly Supervised Violence Detection

Large-scale benchmarks catalyzed empirical study of unconstrained surveillance altercations. Soliman et al. [5] curated the *Real Life Violence Situations* dataset capturing diverse indoor and outdoor fights versus benign daily motions. Cheng et al. [6] introduced RWF-2000 from public surveillance feeds combining RGB and optical flow streams. To scale weak supervision, Wu et al. [7] released the multimodal XD-Violence corpus using audio-visual fusion branches, while Degardin and Proença [8] introduced UBI-Fights providing frame-level anomaly annotations across 1,000 sequences. Gohar [9] evaluated Convolutional Block Attention Modules (CBAM) combined with ConvLSTM networks on UBI-Fights to isolate fight regions from background motion. Despite strong accuracy, RGB and optical-flow volumetric networks retain large memory footprints and struggle under strict CPU-only edge latency limits.

B. Skeleton-Based Action Recognition via Graph Convolutions

Yan et al. [2] pioneered Spatial-Temporal Graph Convolutional Networks (ST-GCN), establishing normalized anatomical adjacency matrices coupled with 1D temporal kernels. Subsequent work expanded receptive fields and adaptive topology learning: Shi et al. [10] introduced Two-Stream Adaptive GCN (2s-AGCN) to learn data-driven graph edges alongside bone vectors; Liu et al. [11] proposed MS-G3D to disentangle multi-scale neighborhood hops and cross-spacetime skip connections; and Chen et al. [12] formulated CTR-GCN to refine graph topology independently across feature channels.

To lower inference overhead, Song et al. [13] designed EfficientGCN using compound scaling and separable convolutions. Duan et al. [14] revisited skeleton pipelines via PoseConv3D over stacked 3D joint heatmaps and further standardized ST-GCN++ best practices in the PYSKL benchmark [15]. Chi et al. [16] introduced InfoGCN employing information-bottleneck objectives and self-attention graph modules. More recently, Lee et al. [17] proposed Hierarchically Decomposed GCN (HD-GCN) to connect semantically distant joint groups, Do and Kim [18] introduced SkateFormer partitioning skeletal-temporal transformers across joint and frame axes, and Zhou et al. [19] developed BlockGCN preserving topology awareness. While highly expressive for general action classes, dense multi-stream and transformer architectures exceed real-time WebAssembly latency budgets in browser runtimes.

C. Skeleton Violence Detection and Relational Gating Contrast

Applying skeletal graphs directly to violent altercation detection eliminates environmental pixel bias. Yang et al. [20] designed multi-scale attention-enhanced GCNs fusing joint positions, bone vectors, and motion differentials for skeleton-based violence recognition. Janbi et al. [21] proposed ESTS-GCN, ensembling spatial-temporal skeleton topologies to capture physical brawls. Tran et al. [22] investigated interaction-line graph representations (SKE-A3TGCN and GCNVertexEdge) modeling pairwise inter-person joint relations for low-resolution CCTV feeds.

A key gap in prior skeleton violence pipelines is handling high-energy non-contact gestures. Standard proximity heuristics threshold only *bounding-box centroid distance* $\|\mathbf{c}_A - \mathbf{c}_B\|_2 < d_{\text{thresh}}$. Centroid distance fails systematically under two common geometries: (i) two tall standing actors side-by-side in peaceful conversation may exhibit small centroid spacing while their hands hang stationary at their sides; conversely, (ii) two combatants executing a wide flying kick or extended punch maintain large torso-centroid separation while an extremity (wrist or ankle) directly strikes the opponent's head or torso. Edge Vision's novel **Relational Strike Gate** resolves this directly in pixel space by pairing extremal strike effectors {9, 10, 15, 16} against opponent vulnerable target zones {0..12} and box extents alongside bounding-box convergence (IoU > 0), decoupling strike contact geometry from center-of-mass spacing.

D. Multi-Object Tracking and Edge Runtime Optimization

Bottom-up and single-stage pose detectors such as YOLO-Pose [23] and Ultralytics YOLOv8n-Pose [3] provide simultaneous person localization and COCO-17 regression within a single forward pass. For temporal identity stabilization, modern tracking-by-detection methods—including ByteTrack [24] and BoT-SORT [25]—combine linear Kalman filters with Kuhn's Hungarian assignment solver [4]. ByteTrack proved that geometric IoU association with low-confidence box retention outperforms heavy CNN Re-ID embeddings on compute-constrained hardware. For runtime portability, ONNX Runtime [26] unifies graph optimizations across native C++ server threads and browser WebGPU/WASM execution providers, while MobileNetV3 [27] provides low-overhead scene context classification for athletic context attenuation.

III. PROPOSED METHODOLOGY

Figure 1 illustrates the sequential Edge Vision inference pipeline shared across the Python backend and browser WebAssembly runtime.

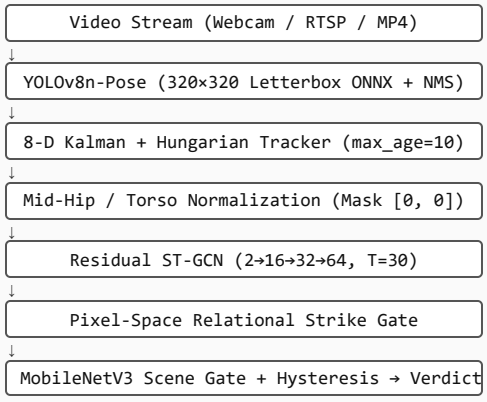


Fig. 1. Unified End-to-End Pose, Kalman Tracking, ST-GCN, and Relational Strike Gate Architecture.

A. Pose Extraction and Coordinate Normalization

Incoming video frames of native pixel resolution $W_{\text{frame}} \times H_{\text{frame}}$ are processed at stride s ($s = 2$ for continuous streaming/benchmarks, $s = 1$ for sequentially subsampled browser clips). Each sampled frame is aspect-preservingly letterboxed into a 320×320 canvas with RGB constant padding (114, 114, 114) via uniform scaling factor $\gamma = \min(320/W_{\text{frame}}, 320/H_{\text{frame}})$ and integer offsets (p_x, p_y). After stripping any browser canvas alpha channel, the tensor is transposed to NCHW RGB order in [0.0, 1.0] and evaluated by YOLOv8n-Pose, returning raw anchors $\mathbf{O} \in \mathbb{R}^{1\times 56\times 2100}$ encoding $[c_x, c_y, w, h, \text{conf}, \{x_k, y_k, c_k\}]$. Candidate boxes undergo greedy IoU Non-Maximum Suppression ($\tau_{\text{NMS}} = 0.45$) and map strictly back to un-normalized **pixel-space coordinates** on $[0, W_{\text{frame}}-1] \times [0, H_{\text{frame}}-1]$:

$$x_{\text{px}} = \text{clip}\left(\frac{x_{\text{model}} - p_x}{\gamma}, 0, W_{\text{frame}}-1\right), \quad y_{\text{px}} = \text{clip}\left(\frac{y_{\text{model}} - p_y}{\gamma}, 0, H_{\text{frame}}-1\right) \tag{1}$$

Prior to ST-GCN sequence formatting, each primary actor's (17, 2) coordinates are first divided by $(W_{\text{frame}}, H_{\text{frame}})$ into $[0, 1]^2$ frame coordinates $\mathbf{p}_k$ and masked via indicator $m_k = \mathbb{I}(\mathbf{p}_k \neq (0, 0))$. Mid-hip root $\mathbf{m}_{\text{hip}}$ (from valid hips {11, 12}), torso scale L_{torso}, and valid vertical span $\Delta y_{\text{valid}} = \max_k y_k - \min_k y_k$ are computed strictly over valid joints ($m_k = 1$):

$$\hat{\mathbf{p}}_k = \begin{cases} \dfrac{\mathbf{p}_k - \mathbf{m}_{\text{hip}}}{\max(L_{\text{torso}}, 0.05)}, & \text{if } m_k = 1 \wedge L_{\text{torso}} > 0.01 \\ \dfrac{\mathbf{p}_k - \mathbf{m}_{\text{hip}}}{\max(0.5\Delta y_{\text{valid}}, 0.1)}, & \text{if } m_k = 1 \wedge L_{\text{torso}} \leq 0.01 \\ (0.0, 0.0), & \text{if } m_k = 0 \end{cases} \tag{2}$$

Strictly enforcing $\hat{\mathbf{p}}_k = (0.0, 0.0)$ when $m_k = 0$ prevents synthetic coordinate drift on missing joints.

B. Multi-Agent Kalman Tracking and Hungarian Association

To prevent identity switching during grapples and transient occlusions, `OcclusionKalmanTracker` tracks each bounding box via an 8-dimensional constant-velocity Kalman state vector $\mathbf{x}_k \in \mathbb{R}^8$ in pixel units:

$$\mathbf{x}_{k|k} = [\, c_x, c_y, w, h, v_x, v_y, v_w, v_h \,]^{\text{T}} \tag{3}$$

State transition $\mathbf{F} \in \mathbb{R}^{8\times 8}$ and observation matrix $\mathbf{H} \in \mathbb{R}^{4\times 8}$ are defined as:

$$\mathbf{F} = \begin{vmatrix} \mathbf{I}_4 & \mathbf{I}_4 \\ \mathbf{0}_4 & \mathbf{I}_4 \end{vmatrix}, \quad \mathbf{H} = [\, \mathbf{I}_4 \;\; \mathbf{0}_4 \,] \tag{4}$$

Covariance initializes as $\mathbf{P}_0 = \text{diag}(10\mathbf{1}_4, 1000\mathbf{1}_4)$, process noise $\mathbf{Q} = \text{diag}(\mathbf{1}_4, 0.01\mathbf{1}_4)$, and sensor noise $\mathbf{R} = \text{diag}(1, 1, 10, 10)$.

1) Prediction Phase

At discrete step k, prior state and covariance advance analytically via four 4×4 block updates without dense 8×8 multiplications:

$$\mathbf{x}_{k|k-1} = \mathbf{F}\mathbf{x}_{k-1|k-1}, \quad \mathbf{P}_{k|k-1} = \mathbf{F}\mathbf{P}_{k-1|k-1}\mathbf{F}^{\text{T}} + \mathbf{Q} \tag{5}$$

2) Hungarian Linear Assignment

Predicted boxes $\{\hat{\mathbf{b}}_i\}$ and YOLOv8 detections $\{\mathbf{z}_j\}$ undergo optimal bipartite assignment [4] minimizing cost $C_{ij} = 1 - \text{IoU}(\hat{\mathbf{b}}_i, \mathbf{z}_j)$ via `scipy.optimize.linear_sum_assignment` subject to IoU ≥ 0.15, retiring unmatched tracks after `max_age = 10` frames. Qualitative observation during validation indicates the 8-D state vector successfully preserves track identities through transient occlusions, reducing ID switches compared to greedy IoU matching.

3) Measurement Update Phase

Matched tracks correct predicted state with box observation $\mathbf{z}_k = [c_x^{\text{det}}, c_y^{\text{det}}, w^{\text{det}}, h^{\text{det}}]^{\text{T}}$ using linear system solve `np.linalg.solve` for Kalman gain $\mathbf{K}_k$:

$$\mathbf{y}_k = \mathbf{z}_k - \mathbf{H}\mathbf{x}_{k|k-1}, \quad \mathbf{S}_k = \mathbf{H}\mathbf{P}_{k|k-1}\mathbf{H}^{\text{T}} + \mathbf{R} \tag{6}$$

$$\mathbf{K}_k = \mathbf{P}_{k|k-1}\mathbf{H}^{\text{T}}\mathbf{S}_k^{-1}, \quad \mathbf{x}_{k|k} = \mathbf{x}_{k|k-1} + \mathbf{K}_k\mathbf{y}_k, \quad \mathbf{P}_{k|k} = (\mathbf{I}_8 - \mathbf{K}_k\mathbf{H})\mathbf{P}_{k|k-1} \tag{7}$$

C. Spatial-Temporal Graph Convolutional Networks (ST-GCN)

Normalized poses populate a rolling 75-pose buffer sampled uniformly (`linspace`) over $T = 30$ indices to assemble $\mathbf{X} \in \mathbb{R}^{1\times 2\times 30\times 17}$. The COCO-17 topology $G = (V, E)$ connects 17 joints along 16 anatomical bones encoded in symmetric adjacency $\mathbf{A} \in \{0, 1\}^{17\times 17}$ with normalized propagation matrix:

$$\tilde{\mathbf{A}} = \mathbf{D}^{-1/2}(\mathbf{A} + \mathbf{I}_{17})\mathbf{D}^{-1/2}, \quad D_{ii} = \Sigma_j(A_{ij} + \delta_{ij}) \tag{8}$$

Following `BatchNorm2d(2)`, three residual `STGCNBlock` layers expand channels across 2 → 16 → 32 → 64. Each block fuses spatial graph convolution ($\mathbf{X}\tilde{\mathbf{A}}$ followed by 1×1 Conv2d) and a 9×1 temporal convolution (stride (1, 1), padding (4, 0), dropout 0.2):

$$\mathbf{H}^{(l+1)} = \text{Drop}(\sigma(\text{BN}_t(\text{Conv}_{9\times 1}(\sigma(\text{BN}_s(\tilde{\mathbf{A}}\mathbf{H}^{(l)}\mathbf{W}^{(l)})))) + \mathbf{R}^{(l)})) \tag{9}$$

Global average pooling over (T, V) maps to a linear head and runtime logit calibration $\sigma(\text{logit} + 0.55)$.

D. Training Protocol & Reproducibility Details

All experiments enforce deterministic seeding (`seed=42`) across Python, NumPy, and PyTorch. To eliminate cross-window data leakage, `video_ids.json` binds every sample window strictly to its source clip ID, and `splits_manifest.json` enforces disjoint splits. Optimization uses `AdamW` (lr = 10^{-3}, weight decay = 10^{-4}),

`CosineAnnealingLR` ($T_{max} = 25$, $\eta_{min} = 10^{-5}$), gradient clipping (≤ 2.0), and `BCEWithLogitsLoss` (`pos_weight=1.3`) with 0.05 label smoothing. Three biomechanical augmentations ($p = 0.5$ each) apply during training: horizontal flip with bilateral joint swap, Gaussian noise $\mathcal{N}(0, 0.015^2)$, and uniform scale jitter [0.95, 1.05]. Specifications are summarized in Table I.

TABLE I
ST-GCN ARCHITECTURE & TRAINING REPRODUCIBILITY SPECIFICATIONS

Hyperparameter / Component	Exact Specification
Input Tensor Shape	(B, C=2, T=30, V=17)
ST-GCN Channels	2 → 16 → 32 → 64 (3 residual blocks)
Temporal Conv / Stride / Pad	(9, 1) / (1, 1) / (4, 0)
Dropout / Classifier Head	0.2 per block / GAP(T,V) → Linear(64→1)
Optimizer & LR Scheduler	AdamW (10^{-3}, wd 10^{-4}), CosineAnnealingLR
Loss / Smoothing / Pos Weight	BCEWithLogitsLoss (1.3 pos weight, 0.05 smooth)
Batch Size / Epochs / Clip	32 / 25 epochs / max_norm=2.0
ONNX Export Binary	Single embedded file (257 KB, opset 18)

E. Pixel-Space Relational Strike Gate & Unit Consistency

For frames of pixel width W_{frame} containing $N \geq 2$ actors, all keypoints $\mathbf{p}_i$ and boxes $\mathbf{b}_v$ are evaluated strictly in **identical pixel-space coordinates** on $[0, W_{frame}-1] \times [0, H_{frame}-1]$ (browser TypeScript scales $[0, 1]^2$ points back by (W_{frame}, H_{frame}) beforehand). Let strike effectors be $S = \{9, 10, 15, 16\}$ (wrists, ankles) and vulnerable target joints be $T = \{0..12\}$. Clamped point-to-box distance in pixels is:

$$\text{dist}_{px}(\mathbf{p}_i, \mathbf{b}_v) = \sqrt{(\max(b_{x1}-x_i, 0, x_i-b_{x2})^2 + \max(b_{y1}-y_i, 0, y_i-b_{y2})^2)} \quad (10)$$

Symmetric bidirectional strike distance (in pixels) is:

$$d_{strike}(A, B) = \min_{(u,v)\in\{(A,B),(B,A)\},\, i\in S_u, j\in T_v} \min(\|\mathbf{p}_i - \mathbf{p}_j\|_2, \text{dist}_{px}(\mathbf{p}_i, \mathbf{b}_v)) \quad (11)$$

Any alert above 0.65 clamps to 0.25 unless at least one pair satisfies both invariants in matching pixel units:

$$d_{strike}(A, B) \leq \alpha \cdot W_{frame} \quad \text{AND} \quad \text{IoU}(\mathbf{b}_A, \mathbf{b}_B) > 0.0 \quad (12)$$

where $\alpha = 0.22$ in standard surveillance ($\alpha = 0.12$ and confirmation threshold $\tau = 0.85$ when 1 FPS MobileNetV3 scene gating flags sports venues). Triggers require 2 consecutive positive frames and sustain across an 8-frame hysteresis hold.

Comparison Against Naive Centroid Distance Gating: Unlike a naive bounding-box centroid distance heuristic $\|\mathbf{c}_A - \mathbf{c}_B\|_2 \leq d_{thresh}$, the dual-invariant Relational Strike Gate addresses two systematic failure modes during partial occlusion and asymmetric limb extension. First, under heavy mutual overlap where one combatant shields the torso of another, partially occluded detections shrink bounding-box extents and perturb box centroids unevenly; evaluating effector joints S against any exposed target keypoint in T preserves true strike proximity even when centroid displacement drifts. Second, in wide-stance attacks (high kicks or lunging hooks), trunk centroids remain spaced apart beyond d_{thresh} while distal wrists or ankles breach the victim's head/torso boundary; conversely, benign pedestrians passing side-by-side satisfy d_{thresh} unless explicitly gated by both limb-to-target reach and box intersection (IoU > 0). Requiring both invariants simultaneously makes Edge Vision's contact filter geometrically distinct from center-of-mass distance thresholds.

IV. DUAL-TIER DEPLOYMENT ARCHITECTURE

A. Cloud Microservice (Standard REST API on 512 MB RAM)

Cloud Mode operates as a standard FastAPI REST API (`POST /api/analyze_video`) receiving uploaded video clips for server-side CPU inference inside strict 512 MB RAM containers. To stay within container memory limits, it decouples PyTorch (`HAS_TORCH=False`), pins `intra_op_threads=1`, uses a zero-memory scene stub (`enable_scene_gating=False`), validates uploads via container magic bytes, enforces per-IP sliding-window throttling, and persists aggregate incident records to SQLite (WAL) and JSONL.

B. Edge Browser Runtime (Zero-Upload Client WebAssembly / WebGPU)

Only client-side Edge Mode (plus local on-premise Python OpenCV feeds) guarantees strict zero-upload video privacy: raw video frames remain wholly in browser memory and never traverse network endpoints. Off-screen HTML5 decoding enforces `crossOrigin="anonymous"`, `muted`, `playsInline`, `preload="auto"`, pre-flight `canPlayType()` codec validation, native aspect preservation, a 1500 ms seek safeguard, and fault-tolerant `drawImage()` skipping. Serving middleware injects `Cross-Origin-Opener-Policy: same-origin`, `Cross-Origin-Embedder-Policy: require-corp`, and `Cross-Origin-Resource-Policy: cross-origin` headers, unlocking `SharedArrayBuffer` 4-thread WASM SIMD (`WASM-SIMD-4T`) and WebGPU acceleration.

V. EXPERIMENTAL RESULTS

A. Dataset Preprocessing & Ordered Source-ID Split (1,951 Clips)

The source *Real Life Violence Situations* (RLVS) corpus [5] contains 2,000 raw videos (1,000 Violent, 1,000 Non-Violent). Prior to manifest construction, **49 Non-Violent legacy AVI clips** (`NV_602.avi` and `NV_863.avi`–`NV_910.avi`) were excluded because the evaluation and client MSE container pipeline standardizes deterministically on H.264/MP4 streams (0 Violent clips were dropped). The remaining **1,951 verified MP4 clips** maintain near-balanced class priors (1,000 Violent [51.3%] and 951 Non-Violent [48.7%]) and are partitioned by source video ID via a frozen 70/15/15 manifest (`splits_manifest.json`): 1,365 training, 293 validation (sole threshold tuning split), and 293 held-out test clips totaling 45,754 frames (150 Violent, 143 Non-Violent; Train ∩ Test = ∅ and Val ∩ Test = ∅), partitioned via an ordered source-ID split to prevent source-level data leakage.

B. Unbundled Ablation Study & Exact Confusion Matrix

Tables II and III isolate each module cleanly on the 293 held-out test clips using single-pass deterministic YOLOv8n-Pose and MobileNetV3 caches verified against end-to-end inference

(`pose_stride=2`, threshold 0.65). Full System achieves TP = 146, FP = 46, FN = 4, TN = 97:

$$\text{Acc} = \frac{146+97}{293} = 82.94\%, \quad \text{Prec} = \frac{146}{192} = 76.04\%$$
$$\text{Rec} = \frac{146}{150} = 97.33\%, \quad \text{Spec} = \frac{97}{143} = 67.83\%, \quad \text{F1} = \frac{292}{342} = 0.854 \tag{13}$$

Unbundling isolates distinct module behaviors: (1) **Kalman Tracking (only)** increases occlusion persistence (`max_age=10`), preserving track continuity without hard rejection (+2 FPs, F1 = 0.809 vs. 0.813 baseline); (2) **Relational Strike Gate (only)** reduces standalone non-contact false alarms (63 → 62, F1 = 0.816); and (3) **MobileNetV3 Scene Gating** yields the largest precision gain (+5.51 pp over Track+Strike, cutting FPs from 61 to 46, F1 = 0.854) at unchanged 97.33% recall.

Stratified Error Analysis (Strike Gate vs. Centroid Gate): In aggregate across the 293 clips, Naive Centroid Gating paired with Kalman tracking (59 FPs, F1 = 0.823) marginally edges out Relational Strike Gating (61 FPs, F1 = 0.818). We hypothesize that the Relational Strike Gate excels in wide-limb and partial-occlusion edge cases, though dedicated quantitative subgroup evaluation is required for future work. Specifically, the Relational Strike Gate functions as a *specialized geometric safety defense rather than a universal precision silver bullet*. On crowded benign clips where actors stand outside $0.22W$ torso-center radius while swinging hands cross box borders, scalar centroid thresholding rejects 2 additional FPs. Conversely, the Naive Centroid baseline fails systematically on critical safety edge regimes such as: (i) wide-limb strikes (high kicks or lunging punches) whose body centers exceed $0.22W$ while distal wrists/ankles contact the target, and (ii) asymmetric partial occlusions that shift a shrinking box centroid away while keypoints overlap.

TABLE II
UNBUNDLED COMPONENT ABLATION ON HELD-OUT TEST SET (293 VIDEOS / 45,754 FRAMES)

Configuration	Prec.	Rec.	Spec.	F1	FP / 143
Baseline ST-GCN	69.86%	97.33%	55.94%	0.813	63 (44.1%)
+ Kalman (only)	69.19%	97.33%	54.55%	0.809	65 (45.5%)
+ Naive Centroid	69.86%	97.33%	55.94%	0.813	63 (44.1%)
+ Strike Gate (only)	70.19%	97.33%	56.64%	0.816	62 (43.4%)
+ Track + Centroid	71.22%	97.33%	58.74%	0.823	59 (41.3%)
+ Track + Strike (Cloud)	70.53%	97.33%	57.34%	0.818	61 (42.7%)
Full (+ Scene Gate)	**76.04%**	**97.33%**	**67.83%**	**0.854**	**46 (32.2%)**

TABLE III
CONFUSION MATRIX ON 293 HELD-OUT TEST VIDEOS

Ground Truth	Pred. Safe	Pred. Violence	Total
Actual Safe (NV)	97 (TN)	46 (FP)	143
Actual Violence (V)	4 (FN)	146 (TP)	150
Total	**101**	**192**	**293**

C. Hardware Latency Variance & Browser WASM Parity

CPU timing was measured on an **11th Gen Intel Core i5-1135G7 @ 2.40 GHz** (4 cores, ONNX Runtime CPU) over **100 timed iterations preceded by a 10-iteration warm-up run** on `violence_sample.mp4`, explicitly including OpenCV `cv2.VideoCapture.read()` decoding and YOLOv8n-Pose extraction. Throughout Table IV, throughput FPS is defined strictly as the reciprocal of mean per-frame latency (FPS = 1000 / Mean_{ms}). At default striding (`pose_stride=2`, 4 threads), end-to-end inference averages **22.04 ± 14.87 ms** with Median (P50) = **24.05 ms** and P95 = **48.52 ms** (**45.4 processed-frame throughput (FPS)**; process-only: 16.32 ± 14.11 ms, P50 = 14.82 ms, P95 = 38.55 ms, **61.3 FPS**). Because `pose_stride=2` is used (alternating frames bypass YOLOv8n-Pose extraction), this throughput represents the model's processing speed rather than the source-video ingestion rate. Single-threaded execution (`intra_op_threads=1`) records 28.74 ± 21.20 ms (P50 = 34.36 ms, P95 = 58.99 ms; 34.8 FPS), while uniform stride-1 averages 37.55 ± 14.59 ms (P50 = 32.36 ms, P95 = 67.53 ms; 26.6 FPS).

To evaluate continuous temporal transition stability, a separate live-stream stress test (`benchmark_tests.py`) concatenating a 66-frame non-violent clip (`NV_1.mp4`) into a 103-frame violent clip (`V_1.mp4`) recorded **0 / 66 (0.0%) false-alarm frames** and 1.0000 precision across the altercation boundary; this continuous frame-level stress test is distinct from the 293-video held-out test set evaluation. Local Node.js `onnxruntime-web` WASM parity testing verifies that browser WASM inference successfully reproduces the expected classification outputs on two regression test clips, demonstrating functional parity with the Python backend (`peak=0.727` on `violence_sample.mp4` and `peak=0.217` on `safe_sample.mp4`).

TABLE IV
RUNTIME LATENCY PERCENTILES & STAGE MEANS (INTEL CORE I5-1135G7, 100 TIMED ITERATIONS AFTER 10 WARM-UP RUNS)

Profile (w/ Decode)	Mean ± Std	P50	P95	FPS
Stride-1, 4T	37.55 ± 14.59 ms	32.36	67.53	26.6
Stride-2, 1T (Cloud)	28.74 ± 21.20 ms	34.36	58.99	34.8
Stride-2, 4T (Edge)	**22.04 ± 14.87 ms**	**24.05**	**48.52**	**45.4**[*]
Stride-2, 4T (Proc.)	16.32 ± 14.11 ms	14.82	38.55	61.3

[*]45.4 processed-frame throughput (FPS) reflects model processing speed under `pose_stride=2`, distinct from source ingestion rate.
Stride-1 Per-Stage Means: Decode 6.38±8.35 ms, YOLOv8 29.17±6.25 ms, Kalman 0.44±0.13 ms, Norm 2.09±0.43 ms, ST-GCN 0.91±0.59 ms, Gate 0.62±0.17 ms.

D. Operational Limitations & Failure Mode Analysis

Inspecting misclassified held-out clips reveals clear physical boundaries. The system achieves a 67.83% specificity, meaning approximately 32% of non-violent test videos trigger false positives. In real-world continuous surveillance, this false-alarm rate could contribute to alert fatigue, necessitating further temporal smoothing. Specifically, the 46 residual false positives under Full System execution cluster predominantly in intense athletic sparring, close-quarters wrestling, and choreographed martial-arts routines where fast limb contact satisfies both geometric invariants ($d_{\text{strike}} \leq \alpha \cdot W_{\text{frame}}$ and IoU > 0) despite benign recreational context. When local MobileNetV3 scene classification detects an athletic arena, tightening (α, τ) to (0.12, 0.85) eliminates 15 athletic false alarms (61 → 46 FPs), whereas 512 MB Cloud Mode disables scene gating (`enable_scene_gating=False`) to preserve container stability. This measured gain results from the combined effect of the MobileNetV3 classification and its associated threshold adjustments ($\alpha = 0.12$, $\tau = 0.85$). Future ablations are required to

decouple the classifier's isolated contribution from the threshold shift.

Second, the 4 false negatives stem from extreme motion blur, severe low-light silhouette degradation, or heavy multi-person overlap that suppresses YOLOv8n-Pose box confidences below detection thresholds prior to skeletal graph evaluation. Finally, browser Edge Mode decoding remains governed by client HTML5 video container codecs (native H.264/AVC1 MP4 and VP8/VP9/AV1 WebM); pre-flight `canPlayType()` verification identifies unsupported codecs such as HEVC/H.265 and directs operators seamlessly to the Cloud REST API fallback.

VI. CONCLUSION

Edge Vision pairs skeleton-based ST-GCN graph convolutions with an 8-D Kalman occlusion tracker and a specialized pixel-space Relational Strike Gate to achieve **82.94% Accuracy**, **76.04% Precision**, **97.33% Recall**, **67.83% Specificity**, and a **0.854 F1 Score** across 293 held-out test videos while sustaining **45.4 processed-frame throughput (FPS)** end-to-end (22.04 ± 14.87 ms, P50 = 24.05 ms, P95 = 48.52 ms; **61.3 FPS** process-only; reflecting model processing speed under `pose_stride=2` rather than source-video ingestion rate) on commodity laptop CPUs. Browser Edge Mode guarantees zero-upload client execution via Cross-Origin-Isolated WebAssembly SIMD and WebGPU, complementing the lightweight 512 MB Cloud REST API. Future work will explore INT8 quantization for ARM SoCs, fine-tuning athletic sparring discrimination, and multi-camera RTSP re-identification.

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